$C_{qd}^{(1)}\mathrm{i1\_i2\_i3\_i4\_} \rightarrow$

$$\frac{1}{16\pi^2}\Big(-\frac{2}{243}\sum_p g_1^4\,LF_{3,0}[m_{\tilde{d}}^p]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{324}\sum_p g_1^4\,LF_{4,-1}[m_{\tilde{d}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{1215}\sum_p g_1^4$$

$$LF_{5,-2}[m_{\tilde{d}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}\sum_p g_1^4\,LF_{3,0}[m_{\tilde{e}}^p]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{108}\sum_p g_1^4\,LF_{4,-1}[m_{\tilde{e}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{8}{405}\sum_p g_1^4\,LF_{5,-2}[m_{\tilde{e}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\sum_p g_1^4\,LF_{3,0}[m_{\tilde{l}}^p]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{216}\sum_p g_1^4\,LF_{4,-1}[m_{\tilde{l}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{405}\sum_p g_1^4\,LF_{5,-2}[m_{\tilde{l}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{243}\sum_p g_1^4\,LF_{3,0}[m_{\tilde{q}}^p]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{648}\sum_p g_1^4\,LF_{4,-1}[m_{\tilde{q}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{1215}\sum_p g_1^4\,LF_{5,-2}[m_{\tilde{q}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{243}\sum_p g_1^4\,LF_{3,0}[m_{\tilde{u}}^p]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{81}\sum_p g_1^4\,LF_{4,-1}[m_{\tilde{u}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{32}{1215}\sum_p g_1^4\,LF_{5,-2}[m_{\tilde{u}}^p]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}g_1^4\,LF_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}g_1^4\,LF_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{8}{405}g_1^4\,LF_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}g_1^4\,LF_{2,1,0}[m_1,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}g_1^4\,LF_{2,2,-1}[m_1,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{486}g_1^4\,LF_{3,1,-1}[m_1,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}g_1^4\,LF_{4,1,-2}[m_1,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}g_1^4\,LF_{2,1,0}[m_1,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}g_1^4\,LF_{2,2,-1}[m_1,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{486}g_1^4\,LF_{3,1,-1}[m_1,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}g_1^4\,LF_{4,1,-2}[m_1,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{3888}g_1^4\,LF_{2,1,0}[m_1,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}g_1^4\,LF_{2,2,-1}[m_1,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}g_1^4\,LF_{3,1,-1}[m_1,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}g_1^4\,LF_{4,1,-2}[m_1,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{3888}g_1^4\,LF_{2,1,0}[m_1,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}g_1^4\,LF_{2,2,-1}[m_1,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}g_1^4\,LF_{3,1,-1}[m_1,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}g_1^4\,LF_{4,1,-2}[m_1,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{144}g_1^2\,g_2^2\,LF_{2,1,0}[m_2,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}g_1^2\,g_2^2\,LF_{2,2,-1}[m_2,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}g_1^2\,g_2^2\,LF_{3,1,-1}[m_2,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}g_1^2\,g_2^2\,LF_{4,1,-2}[m_2,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{144}g_1^2\,g_2^2\,LF_{2,1,0}[m_2,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}g_1^2\,g_2^2\,LF_{2,2,-1}[m_2,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}g_1^2\,g_2^2\,LF_{3,1,-1}[m_2,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}g_1^2\,g_2^2\,LF_{4,1,-2}[m_2,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}g_1^2\,g_3^2\,LF_{2,1,0}[m_3,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{2,2,-1}[m_3,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}g_1^2\,g_3^2\,LF_{3,1,-1}[m_3,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{4,1,-2}[m_3,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}g_1^2\,g_3^2\,LF_{2,1,0}[m_3,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{2,2,-1}[m_3,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}g_1^2\,g_3^2\,LF_{3,1,-1}[m_3,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{4,1,-2}[m_3,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}g_1^2\,g_3^2\,LF_{2,1,0}[m_3,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{2,2,-1}[m_3,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}g_1^2\,g_3^2\,LF_{3,1,-1}[m_3,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{4,1,-2}[m_3,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}g_1^2\,g_3^2\,LF_{2,1,0}[m_3,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{2,2,-1}[m_3,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}g_1^2\,g_3^2\,LF_{3,1,-1}[m_3,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{4,1,-2}[m_3,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{27}g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{2,1,0}[m_{\tilde{d}}^p,\tilde{\mu}]\,\delta_{i3i4}+\frac{1}{54}g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{2,2,-1}[m_{\tilde{d}}^p,\tilde{\mu}]\,\delta_{i3i4}+$$

$$\frac{1}{54}g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{3,1,-1}[m_{\tilde{d}}^p,\tilde{\mu}]\,\delta_{i3i4}+\frac{1}{486}g_1^4\,LF_{2,1,0}[m_{\tilde{d}}^{i3},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}g_1^4\,LF_{3,1,-1}[m_{\tilde{d}}^{i3},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}g_1^2\,g_3^2\,LF_{2,1,0}[m_{\tilde{d}}^{i3},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}g_1^2\,g_3^2\,LF_{3,1,-1}[m_{\tilde{d}}^{i3},m_3]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{486}g_1^4\,LF_{2,1,0}[m_{\tilde{d}}^{i4},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}g_1^4\,LF_{3,1,-1}[m_{\tilde{d}}^{i4},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}g_1^2\,g_3^2\,LF_{2,1,0}[m_{\tilde{d}}^{i4},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}g_1^2\,g_3^2\,LF_{3,1,-1}[m_{\tilde{d}}^{i4},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{6}\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{3,1,0}[m_{\tilde{l}}^p,m_{\tilde{e}}^r]+$$

$$\frac{1}{2}\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{4,1,-1}[m_{\tilde{l}}^p,m_{\tilde{e}}^r]-\frac{1}{3}\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,LF_{5,1,-2}[m_{\tilde{l}}^p,m_{\tilde{e}}^r]-$$

$$\frac{1}{2}\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{3,1,0}[m_{\tilde{q}}^p,m_{\tilde{d}}^r]+\frac{3}{2}\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{4,1,-1}[m_{\tilde{q}}^p,m_{\tilde{d}}^r]-$$

$$\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,LF_{5,1,-2}[m_{\tilde{q}}^p,m_{\tilde{d}}^r]-\frac{1}{54}g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{2,1,0}[m_{\tilde{q}}^p,\tilde{\mu}]\,\delta_{i1i2}+$$

$$\frac{1}{108}g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{2,2,-1}[m_{\tilde{q}}^p,\tilde{\mu}]\,\delta_{i1i2}+\frac{1}{108}g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{3,1,-1}[m_{\tilde{q}}^p,\tilde{\mu}]\,\delta_{i1i2}+$$

$$\frac{1}{1944}g_1^4\,LF_{2,1,0}[m_{\tilde{q}}^{i1},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}g_1^4\,LF_{3,1,-1}[m_{\tilde{q}}^{i1},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}g_1^2\,g_2^2\,LF_{2,1,0}[m_{\tilde{q}}^{i1},m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}g_1^2\,g_2^2\,LF_{3,1,-1}[m_{\tilde{q}}^{i1},m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}g_1^2\,g_3^2\,LF_{2,1,0}[m_{\tilde{q}}^{i1},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{3,1,-1}[m_{\tilde{q}}^{i1},m_3]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}g_1^4\,LF_{2,1,0}[m_{\tilde{q}}^{i2},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}g_1^4\,LF_{3,1,-1}[m_{\tilde{q}}^{i2},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}g_1^2\,g_2^2\,LF_{2,1,0}[m_{\tilde{q}}^{i2},m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}g_1^2\,g_2^2\,LF_{3,1,-1}[m_{\tilde{q}}^{i2},m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}g_1^2\,g_3^2\,LF_{2,1,0}[m_{\tilde{q}}^{i2},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}g_1^2\,g_3^2\,LF_{3,1,-1}[m_{\tilde{q}}^{i2},m_3]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{27}g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{2,1,0}[m_{\tilde{u}}^p,\tilde{\mu}]\,\delta_{i3i4}-\frac{1}{27}g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{2,2,-1}[m_{\tilde{u}}^p,\tilde{\mu}]\,\delta_{i3i4}-$$

$$\frac{1}{27}g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{3,1,-1}[m_{\tilde{u}}^p,\tilde{\mu}]\,\delta_{i3i4}-\frac{1}{2}\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{3,1,0}[m_{\tilde{u}}^r,m_{\tilde{q}}^p]+$$

$$\frac{3}{2}\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{4,1,-1}[m_{\tilde{u}}^r,m_{\tilde{q}}^p]-\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,LF_{5,1,-2}[m_{\tilde{u}}^r,m_{\tilde{q}}^p]+$$

$$\frac{1}{3}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{3,1,-1}[\tilde{\mu},m_1]-\frac{2}{3}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{4,1,-2}[\tilde{\mu},m_1]+$$

$$\frac{1}{3}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{5,1,-3}[\tilde{\mu},m_1]+g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{3,1,-1}[\tilde{\mu},m_2]-$$

$$2\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{4,1,-2}[\tilde{\mu},m_2]+g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{5,1,-3}[\tilde{\mu},m_2]+$$

$$\frac{1}{54}g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{2,1,0}[\tilde{\mu},m_{\tilde{d}}^p]\,\delta_{i3i4}+\frac{5}{108}g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{3,1,-1}[\tilde{\mu},m_{\tilde{d}}^p]\,\delta_{i3i4}-$$

$$\frac{1}{108}g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,LF_{4,1,-2}[\tilde{\mu},m_{\tilde{d}}^p]\,\delta_{i3i4}+\frac{1}{108}g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{2,1,0}[\tilde{\mu},m_{\tilde{q}}^p]\,\delta_{i1i2}+$$

$$\frac{7}{108}g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{3,1,-1}[\tilde{\mu},m_{\tilde{q}}^p]\,\delta_{i1i2}-\frac{1}{54}g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,LF_{4,1,-2}[\tilde{\mu},m_{\tilde{q}}^p]\,\delta_{i1i2}-$$

$$\frac{1}{27}g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{2,1,0}[\tilde{\mu},m_{\tilde{u}}^p]\,\delta_{i3i4}-\frac{1}{108}g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{3,1,-1}[\tilde{\mu},m_{\tilde{u}}^p]\,\delta_{i3i4}-$$

$$\frac{1}{108}g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,LF_{4,1,-2}[\tilde{\mu},m_{\tilde{u}}^p]\,\delta_{i3i4}-\frac{1}{648}g_1^4\,LF_{2,1,1,-1}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{324}g_1^4\,m_1^2\,LF_{2,1,1,0}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{216}m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$LF_{2,2,1,-1}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i2}]-\frac{1}{216}m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,2,1,-1}[m_1,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{648}g_1^4\,LF_{2,1,1,-1}[m_1,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{324}g_1^4\,m_1^2\,LF_{2,1,1,0}[m_1,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{216}m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,2,1,-1}[m_1,m_{\tilde{q}}^{i1},m_{\tilde{d}}^{i4}]-$$

$$\frac{1}{216}m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,2,1,-1}[m_1,m_{\tilde{q}}^{i2},m_{\tilde{d}}^{i3}]-\frac{1}{36}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{d}}^{i3}]-$$

$$\frac{1}{36}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{d}}^{i4}]-\frac{1}{72}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{72}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{q}}^{i2}]-\frac{1}{8}g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}[m_2,\tilde{\mu},m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{8}g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{2,2,1,-2}[m_2,\tilde{\mu},m_{\tilde{q}}^{i2}]-\frac{1}{9}g_3^4\,LF_{2,1,1,-1}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{9}g_3^4\,m_3^2\,LF_{2,1,1,0}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{9}m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,2,1,-1}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i2}]+$$

$$\frac{1}{9}m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,2,1,-1}[m_3,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i1}]-\frac{1}{9}g_3^4\,LF_{2,1,1,-1}[m_3,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{9}g_3^4\,m_3^2\,LF_{2,1,1,0}[m_3,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{9}m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,2,1,-1}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{d}}^{i4}]+$$

$$\frac{1}{9}m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,2,1,-1}[m_3,m_{\tilde{q}}^{i2},m_{\tilde{d}}^{i3}]+\frac{1}{108}m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$LF_{2,1,1,0}[m_{\tilde{d}}^{i3},m_1,m_{\tilde{q}}^{i2}]-\frac{1}{108}m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{3,1,1,-1}[m_{\tilde{d}}^{i3},m_1,m_{\tilde{q}}^{i2}]-$$

$$\frac{2}{9}m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,1,1,0}[m_{\tilde{d}}^{i3},m_3,m_{\tilde{q}}^{i2}]+\frac{2}{9}m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$LF_{3,1,1,-1}[m_{\tilde{d}}^{i3},m_3,m_{\tilde{q}}^{i2}]+\frac{1}{108}m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,1,1,0}[m_{\tilde{d}}^{i4},m_1,m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{108}m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{3,1,1,-1}[m_{\tilde{d}}^{i4},m_1,m_{\tilde{q}}^{i1}]-\frac{2}{9}m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}$$

$$LF_{2,1,1,0}[m_{\tilde{d}}^{i4},m_3,m_{\tilde{q}}^{i1}]+\frac{2}{9}m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{3,1,1,-1}[m_{\tilde{d}}^{i4},m_3,m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{12}\tilde{\mu}^2\,\big(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\big)\,LF_{2,1,1,0}[m_{\tilde{q}}^p,m_{\tilde{u}}^r,\tilde{\mu}]+$$

$$\frac{1}{12}\tilde{\mu}^2\,\big(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\big)\,LF_{3,1,1,-1}[m_{\tilde{q}}^p,m_{\tilde{u}}^r,\tilde{\mu}]+$$

$$\frac{1}{24}\tilde{\mu}^2\,\big(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\big)\,LF_{2,2,1,-1}[m_{\tilde{q}}^p,\tilde{\mu},m_{\tilde{u}}^r]+$$

$$\frac{1}{108}m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,1,1,0}[m_{\tilde{q}}^{i1},m_1,m_{\tilde{d}}^{i4}]-$$

$$\frac{1}{108}m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{3,1,1,-1}[m_{\tilde{q}}^{i1},m_1,m_{\tilde{d}}^{i4}]-$$

$$\frac{2}{9}m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,LF_{2,1,1,0}[m_{\tilde{q}}^{i1},m_3,m_{\tilde{d}}^{i4}]+\frac{2}{9}m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}$$

$$LF_{3,1,1,-1}[m_{\tilde{q}}^{i1},m_3,m_{\tilde{d}}^{i4}]+\frac{1}{108}m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{2,1,1,0}[m_{\tilde{q}}^{i2},m_1,m_{\tilde{d}}^{i3}]-$$

$$\frac{1}{108}m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{3,1,1,-1}[m_{\tilde{q}}^{i2},m_1,m_{\tilde{d}}^{i3}]-\frac{2}{9}m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$LF_{2,1,1,0}[m_{\tilde{q}}^{i2},m_3,m_{\tilde{d}}^{i3}]+\frac{2}{9}m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,LF_{3,1,1,-1}[m_{\tilde{q}}^{i2},m_3,m_{\tilde{d}}^{i3}]-$$

$$\frac{1}{12}\tilde{\mu}^2\,\big(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\big)\,LF_{2,1,1,0}[m_{\tilde{u}}^r,m_{\tilde{q}}^p,\tilde{\mu}]+$$

$$\frac{1}{12}\tilde{\mu}^2\,\big(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\big)\,LF_{3,1,1,-1}[m_{\tilde{u}}^r,m_{\tilde{q}}^p,\tilde{\mu}]+$$

$$\frac{1}{24}\tilde{\mu}^2\,\big(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r}\big)\,LF_{2,2,1,-1}[m_{\tilde{u}}^r,\tilde{\mu},m_{\tilde{q}}^p]-$$

$$\frac{1}{4}\overline{y_d}^{ri3}\,\overline{y_d}^{i2p}\,y_d^{ri4}\,y_d^{i1p}\,LF_{2,1,1,-1}[\tilde{\mu},m_{\tilde{d}}^p,m_{\tilde{q}}^r]-$$

$$\frac{1}{2}\tilde{\mu}^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\overline{y_u}^{i2r}\,y_u^{i1r}\,LF_{2,1,1,0}[\tilde{\mu},m_{\tilde{q}}^p,m_{\tilde{u}}^r]+$$

$$\frac{1}{54}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{d}}^{i4},\tilde{\mu}]+\frac{1}{108}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$LF_{1,1,1,1,-1}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1},\tilde{\mu}]+\frac{1}{108}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}[m_1,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2},\tilde{\mu}]+$$

$$\frac{1}{216}g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}[m_1,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2},\tilde{\mu}]+\frac{1}{8}g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$LF_{1,1,1,1,-1}[m_2,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2},\tilde{\mu}]+\frac{2}{9}g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{d}}^{i4},\tilde{\mu}]-$$

$$\frac{2}{9}g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1},\tilde{\mu}]-\frac{2}{9}g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$LF_{1,1,1,1,-1}[m_3,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2},\tilde{\mu}]+\frac{2}{9}g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,LF_{1,1,1,1,-1}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2},\tilde{\mu}]\Big)$$